

1 Reliable comparison of severe felt-intensity
2 classifiers with versioned and dependent event
3 data: a USGS DYFI benchmark

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9 **Abstract**

10 Engineers comparing machine-learning classifiers on evolving event catalogues
11 can reach false conclusions when dependent events cross data splits or small score
12 differences are treated as wins. We ask whether candidate severe felt-intensity
13 classifiers comply with predefined rules for allowed inputs and evaluation, and
14 whether their apparent improvements remain supported after catalogue version,
15 event grouping, temporal evaluation, and paired uncertainty are fixed. Using a
16 frozen USGS “Did You Feel It?” (DYFI) snapshot of 7,520 $M \geq 5$ records, we
17 formed 2,362 eligible events, reserved 430 events from 2023–2024 as a tempo-
18 ral holdout, and evaluated fixed baselines using Brier score and sequence-cluster
19 bootstrap intervals. Four learned models improved on the no-skill reference, but
20 the two lowest-Brier models remained unresolved; one candidate violated the
21 allowed-input rule by using event year. For a fixed random-forest diagnostic,
22 sequence grouping changed Brier score by -0.006 and area under the receiver-
23 operating-characteristic curve (AUROC) by $+0.012$ relative to random folds;
24 because no interval was retained, this contrast is descriptive only. The bench-
25 mark helps engineers reject inadmissible comparisons and unsupported winner
26 claims. Because its labels are public, confirmation requires fresh hidden data or
27 an independent service, and it does not establish real-time warning or operational
28 triage.

Keywords: model comparison, evaluation reliability, benchmark qualification,
dependent-event data, reproducibility, USGS DYFI

1 Introduction

Engineering model selection depends on comparisons that remain valid when a dataset is updated, observations are dependent, and leading scores are close. Without those controls, an apparent improvement can be caused by a changed population, a proxy for the outcome or evaluation axis, cross-split dependence, or unquantified sampling variation rather than by a better model. Standard benchmarking is valuable precisely because it fixes a common task and metrics; recent work in this journal, for example, compares object detectors on COCO to support model selection for engineering applications [29]. The reliability problem is harder for event catalogues because both the catalogue and the dependence structure can evolve.

The U.S. Geological Survey “Did You Feel It?” (DYFI) system is a useful engineering case study. It converts crowdsourced felt reports into a community-decimal intensity (CDI) for each earthquake and is widely reused as an interpreted data product [2, 22, 34, 35]. The target studied here is retrospective classification of whether event-maximum CDI reaches 6 from outcome-blind source metadata. It is not a real-time alert, causal model, or operational triage rule.

Our engineering question is: *do performance and model-superiority claims remain supportable when catalogue version, dependent-event splitting, temporal evaluation, paired uncertainty, and reconstruction are controlled?* The intended users are applied machine-learning engineers evaluating a candidate classifier, maintainers publishing a stable benchmark, and reviewers auditing a claimed improvement. They need two decisions: whether a candidate is admissible for comparison and whether its improvement over a declared reference is supported rather than a point-score fluctuation.

Existing felt-intensity studies use random or stratified partitions and do not provide that reusable decision contract [15, 20].

1.1 Contributions

We address unreliable model comparison by releasing a versioned benchmark and an executable candidate-qualification contract. Concretely, the work provides:

1. **A single-source, version-frozen DYFI benchmark** for the task “outcome-blind source metadata $\rightarrow$ later event-maximum $\max\text{CDI} \geq 6$ ”, frozen to one dated access instant so that any later catalogue revision defines a new version rather than a silent overwrite (Section 3).
2. **An executable six-category leakage audit** that rejects direct-label, post-outcome-ascertainment, identifier-proxy, and downstream-product features, and rejects cross-role sequence straddles and temporal-cutoff violations, accepting *only* the exact source-only feature set; it is exercised by 110 unit tests and rejects every deliberately contaminated test case (Section 4.1).
3. **Internally specified, outcome-independent roles** built on order-stable space-time sequence grouping and a temporal external holdout, so that split membership never uses the label (Section 4.2).
4. **A fixed, non-selective baseline suite** with calibration-aware proper scoring, Brier-score sequence-cluster bootstrap uncertainty, explicit handling of ties and unavailable calibration estimates, and single-use authorization of the decision-bearing protected evaluation (Sections 4.4–4.5).
5. **A candidate-qualification workflow** that fails closed on forbidden features or invalid roles and supports an improvement only when paired Brier-score uncertainty excludes zero (Section 5.5).

78 **6. A quantified split-sensitivity diagnostic** (naive random versus sequence-
79 grouped splitting), together with a reproducible release bundle and machine-
80 readable verification record (Section 5).

81 The learned models are worked candidates, not proposed production systems.
82 Their comparison demonstrates the decisions the contract permits: all admissible
83 learned candidates improve over no-skill, one candidate is outside the declared
84 source-only schema, and the two lowest-Brier models remain unresolved.

85 **1.2 Engineering significance and intended use**

86 The output is an engineering test harness for transparent retrospective model compar-
87 ison. A declared candidate enters through the released schema, is excluded if it uses
88 forbidden outcome-derived fields or violates role isolation, is scored on the fixed tempo-
89 ral holdout, and is compared with paired sequence-cluster uncertainty. This workflow
90 prevents an engineer or reviewer from declaring superiority from a different catalogue
91 version, a contaminated split, or an unresolved point difference. Because holdout labels
92 are public, the release is not an untouched prospective test: confirmatory evaluation of
93 a new model requires a fresh benchmark version or hidden evaluation service. Table 1
94 names the failure prevented by each control, and Fig. 1 shows the decision path.

95 **2 Related work and novelty**

96 *DYFI and macroseismic intensity.*

97 The DYFI system and its CDI construction, outlier handling, and post-event revision
98 behaviour are documented in the foundational and review literature [2, 22, 34, 35];
99 these establish that CDI is an after-the-event, response-driven quantity subject to
100 ascertainment effects, which is precisely why our protocol uses response count only to
101 determine eligibility and forbids it as a predictive feature (Section 3). Rösler et al. [24]
102 further show specifically for DYFI that interpreted products can change and should

Table 1 Engineering failure modes addressed by the benchmark qualification contract

Failure source	Invalid engineering conclusion	Qualification control
Catalogue revision	Non-identical event population	Frozen access instant + SHA-256 release
Outcome-derived field	Label leakage	F1/F2 schema audit
Identifier or downstream proxy	Proxy leakage	F3/F4 field audit
Sequence crosses roles	Dependent-event contamination	F5 sequence-role audit
Temporal-cutoff violation	Future information in development	F6 temporal audit
Point-score ranking	False model-superiority claim	Paired sequence-cluster interval
Environment drift	Irreproducible scores/artifacts	Pinned environment + reconstruction hash

103 be preserved with timestamps for reproducible analysis. The dated, hashed snapshot
 104 released here implements that preservation principle. We reuse these provenance facts
 105 rather than re-deriving them.

106 *Domain benchmark resources.*

107 The INGe resource links Italian macroseismic intensities to colocated strong-motion
 108 records for benchmarking intensity-ground-motion relationships [18]. It is a domain-
 109 near resource precedent, but differs from the present event-level USGS DYFI
 110 classification task, source-only feature boundary, leakage audit, and protected tem-
 111 poral evaluation. The 2026 USGS Western North America resource contains about
 112 696,000 site-level macroseismic observations from about 5,600 earthquakes and is pre-
 113 dominantly DYFI [6]. It is a larger observational resource, but does not define this
 114 event-maximum binary task, executable leakage audit, or frozen temporal evaluation.
 115 Stockman et al. [28] provide the nearest broader leakage-aware seismological bench-
 116 mark precedent: EarthquakeNPP standardizes public catalogues, repairs an improper
 117 earthquake-forecasting split, compares neural point processes with ETAS, and finds

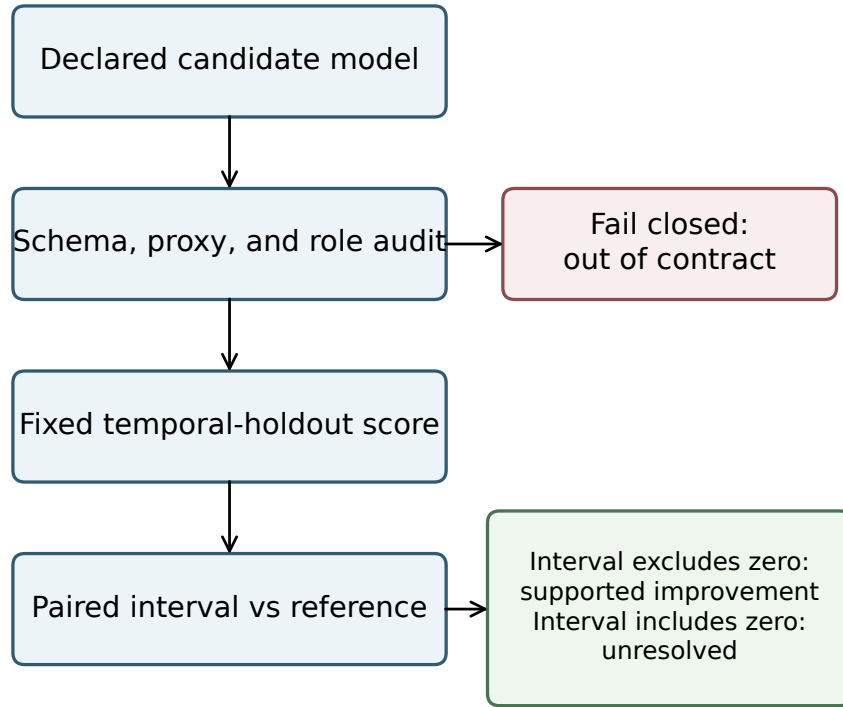


Fig. 1 Candidate-model qualification. A candidate that fails the schema, proxy, role, or temporal audit is inadmissible. An admitted candidate is scored on the fixed holdout and compared with paired uncertainty; an interval including zero remains unresolved rather than becoming a winner claim

no consistent neural-model advantage. Its target is occurrence forecasting with chronological likelihood/CSEP evaluation, rather than source metadata predicting a later citizen-reported felt-intensity endpoint; the resources are therefore methodologically adjacent but task-distinct.

Engineering benchmark and model-comparison precedents.

Comparative engineering studies use a fixed task and common metrics to turn model outputs into selection evidence. Tian et al. [29], for example, compare object detectors on COCO to support model choice for smart-city applications. In this journal,

126 machine-learning comparisons have also been tied to explicit engineering decisions in
127 rock-property estimation and pavement maintenance [1, 17], while seismic engineering
128 studies state the response quantity and design interpretation affected by the analy-
129 sis [26]. The present contribution follows that decision-oriented form but evaluates
130 the reliability of the comparison itself: whether a candidate is admissible, whether it
131 improves over a reference, and whether the evidence supports naming a winner.

132 An adjacent event-level binary study used post-event LastQuake felt reports to
133 classify high- versus low-impact earthquakes and evaluated later 2022 events [14]. Its
134 crowdsourced predictors, impact endpoint, and rapid-response objective differ from
135 the source-only DYFI benchmark and reusable evaluation contract developed here.

136 *Macroseismic databases and citizen observations.*

137 Event-specific analysis in the 2019 Athens earthquake combined online felt reports
138 with ground motion and source modelling [12]. Crowdsourced macroseismic observa-
139 tions have also been used to characterize geological settings and spatial variations in
140 shaking response [25]. Curated macroseismic resources such as the Belgian traditional
141 macroseismic database and MACROSISSDATA preserve historical intensity observa-
142 tions for transparent reuse and later analysis [16, 27]. The present work extends this
143 seismological resource tradition from data preservation to a versioned, executable
144 contract for model evaluation.

145 *Machine learning on felt/macroseismic intensity.*

146 Recent studies predict macroseismic intensity with random forests and gradient boost-
147 ing from instrumental or catalogue features [15, 20]. Machine learning has also been
148 applied to earthquake alert-level classification using intensity-related features and to
149 waveform detection with openly released seismic datasets [5, 19]. Those tasks use post-
150 event intensity descriptors or waveforms, rather than the source-only feature boundary
151 studied here. Each felt-intensity model application is evaluated under a random or

152 stratified partition and does not release a reusable, event-grouped, leakage-audited
 153 benchmark; Mori and Naso [15] moreover predict from instrumental ground-motion
 154 parameters rather than outcome-blind catalogue metadata, and Patelli et al. [20]
 155 evaluate a single-system train/test split on seismometric inputs. Their concrete draw-
 156 back for our purpose is the evaluation contract: a random or stratified split does not
 157 remove within-sequence near-duplicate contamination, and none ships an executable
 158 leakage audit or a version-frozen resource that a third party can rerun. Our protocol
 159 addresses that specific drawback with internally specified sequence-grouped roles and
 160 an executable six-category audit.

161 *DYFI-based intensity models and concurrent leakage-aware work.*

162 Closest to our exact data source, Karthik Kumar et al. [10] predict DYFI commu-
 163 nity intensity from the same catalogue metadata (magnitude, depth, location) with a
 164 random-forest regressor on Indian events under a single naive train/test split (reported
 165 $R^2 = 0.829$); our resource does not fit a competing regressor but instead supplies
 166 a versioned, digitally fingerprinted, event- and sequence-grouped benchmark with
 167 an executable leakage audit and a protected decision-bearing evaluation authorized
 168 once, in place of the naive split that setup uses. Concurrently, Li et al. [13] release
 169 a DYFI-grounded benchmark that simulates Modified Mercalli Intensity (MMI) for
 170 two earthquakes (2014 Napa and 2019 Ridgecrest) with large language models and
 171 includes an explicit location-identifier leakage test; our resource differs in task and
 172 construction—an outcome-blind source-metadata→later-severe binary classification
 173 over a decade-scale $M \geq 5$ corpus with internally specified sequence-grouped roles,
 174 rather than multimodal per-zip-code MMI simulation for two events—so the two are
 175 complementary rather than duplicative.

176 *Cross-validation for dependent data.*

177 That grouped or blocked cross-validation is the correct evaluation strategy for tem-
178 porally, spatially, or hierarchically structured data is established practice [11, 23]; we
179 do not claim this as novel.

180 *Novelty delta.*

181 Conceding the above, the delta this resource supplies that prior work does not is the
182 *released, versioned, hashed, single-source DYFI benchmark* itself: an executable six-
183 category audit covering forbidden fields, proxy variables, and split contamination; an
184 internally specified, sequence-grouped role assignment with a temporal external hold-
185 out, a documented severe endpoint with threshold and threshold sensitivity, a fixed
186 non-selective baseline suite with a no-skill comparator, a calibration/uncertainty/tie
187 contract, an exact-reproduction verification, and a quantified naive-versus-grouped
188 split diagnostic. A current-year search located no released resource occupying this
189 exact combination; novelty is stated as a concrete delta to located work, with no
190 first/only superlative asserted.

191 **3 Data, provenance, and outcome-boundary** 192 **disclosure**

193 *Source and single-instant pinning.*

194 The corpus is derived entirely from USGS DYFI and its parent ANSS Comprehensive
195 Catalog (ComCat), retrieved through the International Federation of Digital Seis-
196 mograph Networks (FDSN) event web service [30, 33]. The acquisition envelope is
197 outcome-independent: $M \geq 5$, product type `dyfi`, event type `earthquake`, starttime
198 2015-01-01 to endtime 2025-01-01. FDSN defines the end bound as inclusive; no frozen
199 record occurs at that boundary, so the realized snapshot spans 2015–2024. Results are
200 ordered by time. The preliminary count request and the final data request use this

201 same envelope. Because the catalogue evolves, the downloaded data were frozen at
 202 one access instant (2026-09-01T06:33:30Z) and assigned a SHA-256 digital fingerprint.
 203 A preliminary count changed from 7,575 to 7,576 during the acquisition session, doc-
 204 umenting one live-catalogue revision. The final downloaded snapshot contained 7,520
 205 DYFI records, and a count requested 50 seconds later in the same acquisition session
 206 also returned 7,520. Thus, the one-record change belongs only to the earlier prelim-
 207 inary checks; it is not a discrepancy in the frozen analysis snapshot. Likewise, the
 208 56-record difference between the last preliminary count and the final 7,520-record snap-
 209 shot is not an analysis exclusion: the preliminary count had no paired payload, so its
 210 membership difference cannot be decomposed and is retained only as catalogue-version
 211 provenance. The near-contemporaneous final query and count agree exactly. The dig-
 212 ital fingerprint of the downloaded bytes, rather than a preliminary count, defines this
 213 resource version. The frozen snapshot contains 7,520 DYFI records (Table 2, Fig. 2).

214 ***Protocol chronology.***

215 The file records an internal 2026-09-01 freeze but was first publicly timestamped on
 216 2026-09-19. We therefore describe it as an internally frozen protocol, not an externally
 217 registered preregistration.

218 ***Retrospective eligibility and post-report ascertainment.***

219 DYFI intensity is an after-the-event quantity whose reliability grows with the number
 220 of felt reports [22]. To keep the label meaningful we apply a data-informed eligibility
 221 filter of at least ten responses per event. The response count is used *only* for eligibil-
 222 ity and is explicitly forbidden as a predictive feature, because it is a post-outcome
 223 ascertainment proxy; this forbiddance is enforced by the leakage audit (family F2,
 224 Section 4.1). Eligibility yields 2,362 events with severe prevalence 0.298 (703 severe
 225 events). Of these, 2,360 events are assigned to the internally specified analysis roles
 226 (Section 4.2) and contain 702 severe events; the remaining two are temporal-straddle

227 exclusions, one of which is severe. That single severe straddle event accounts for the
228 difference between the eligible cohort (703 severe) and the analysis-assigned set (702
229 severe), so eligibility is not the same as analysis-role assignment (Table 2).

230 ***Endpoint.***

231 We label an event severe when its event-maximum CDI is at least 6.0 (intensity VI or
232 greater). We use 6.0 as the primary threshold and report sensitivity at 5.5, 6.0, and
233 6.5. We also examine a near-threshold band of ± 0.5 (Section 5.2).

234 ***Outcome-boundary disclosure and held-out semantics.***

235 Only corpus-level outcome-blind counts and inherited aggregate feasibility statistics
236 were exposed before construction; no per-event label informed the endpoint, split, or
237 tuning choices, and role membership is assigned by an internally specified, outcome-
238 independent rule that never uses the label. We therefore describe the external holdout
239 as *held out from model development with aggregate feasibility previously known*, and we
240 deliberately do *not* call it “untouched” or “confirmatory.” Two event identifiers were
241 prelisted for quarantine because their aggregate counts had been seen during source
242 verification; neither identifier occurs in the frozen snapshot, so the rule is non-operative
243 in version 1 and excludes no record. The complete list of protocol decisions is reported
244 in Section 5.8. No respondent-level personally identifying information is redistributed;
245 only event-level source metadata, the severe label, the eligibility-only response count,
246 identifiers, and role assignments are released. Snapshot-level provenance is supplied
247 by the activated source-locator manifest and source payload SHA-256; version 1 does
248 not claim per-event product-version hashes. The release bundle follows datasheet-
249 for-datasets documentation [8] and findable, accessible, interoperable, and reusable
250 (FAIR) data principles [36].

251 4 Benchmark design

252 4.1 Feature sets and the six-category leakage audit

253 The primary *source-only* feature set contains magnitude and depth. It is the only
254 feature set evaluated on the temporal holdout. An *expanded-metadata* sensitivity anal-
255 ysis (baseline B3) additionally uses latitude, longitude, and event year. Event year is
256 known at prediction time and is not intrinsically label leakage, but it is excluded by the
257 declared source-only temporal-holdout contract. The audit accepts only an approved
258 feature set with the expected fields in the expected order; any extra or unknown field
259 is rejected.

260 The executable audit enforces six leakage families. Field-level: F1 direct label or
261 its constituents; F2 post-outcome ascertainment or revision (including the response
262 count, eligibility-only); F3 identifier or source proxy; F4 downstream-product proxy.
263 Split-level: F5 sequence cross-role straddle; F6 temporal-cutoff violation. An axis-
264 variable check additionally rejects origin year under a temporal holdout and, when a
265 geographic holdout is configured, latitude/longitude/region. The latter audit branch
266 is provided for reuse but was not executed as an external evaluation in this study.
267 This enforces a conservative B3 scope restriction; it is not evidence that event year
268 is a leaking label proxy. The audit is useful independently of model performance: it
269 must reject every deliberately contaminated test case and accept only the approved
270 clean feature set. It is exercised by 110 unit tests across four suites, all passing, and
271 accepted the source-only feature set in the frozen snapshot.

272 4.2 Sequence grouping and outcome-independent roles

273 Events are grouped into seismic *sequences* by order-stable single-link space-time
274 connected components (100 km, 30 days); the sequence, not the event, is the unit
275 of role assignment, so events linked by this fixed rule cannot straddle roles. This
276 rule is a deterministic grouping heuristic, not a validated catalogue of all physical

277 mainshock–aftershock relationships; unlinked true aftershocks may therefore remain
 278 in different roles. Roles are assigned by an internally specified, outcome-independent
 279 rule (fixed assignment key, temporal cutoff year 2023, development fraction 0.2, five
 280 cross-validation folds); the rule provably never reads the label. The external holdout
 281 contains events on or after 2023. The frozen internal protocol also named a secondary
 282 leave-region-out geographic evaluation, but that branch was not executed and no
 283 geographic role manifest is claimed. The reported geographic result instead summa-
 284 rizes the temporal-holdout predictions by 10° cell (Table 9); Table 8 records this
 285 deviation. Digital fingerprints of the completed temporal role assignments are pub-
 286 lished so that third parties can verify them. The partition is exhaustive and disjoint:
 287 $1,573 + 357 + 430 + 2 = 2,362$ events, comprising 1,573 train, 357 development, 430
 288 external temporal holdout, and 2 temporal-straddle exclusions (Table 2, Fig. 2). Con-
 289 struction (train plus development) is 1,930 events across 1,530 sequences; the holdout
 290 is 430 events across 354 sequences.

291 4.3 Baseline suite and exclusions

292 The fixed suite is B0 no-skill (constant prevalence), B1 magnitude-only logistic regres-
 293 sion, B2 source-only logistic regression, B3 expanded-metadata logistic regression
 294 (used only in compatible sensitivity analyses), B4 random forest (scikit-learn; 21), and
 295 B5 XGBoost [4]. Each learned baseline is tuned *only* by the protocol-defined deter-
 296 ministic grouped cross-validation on the construction corpus over a fixed grid, named
 297 before evaluation, and carried unchanged to the protected holdout with frozen best
 298 parameters. A machine-readable fit log enforces a maximum of 120 model fits; the exe-
 299 cuted run used 91. Two comparators are excluded for protocol-defined, reproducible
 300 reasons recorded in the protocol-decision table (Section 5.8): B3 is not evaluated
 301 on the temporal holdout because origin year is outside the declared source-only fea-
 302 ture contract; and the Atkinson–Wald (2007) DYFI intensity-prediction equation [2]

303 is excluded because it cannot be applied to the source-only feature set and tectoni-
304 cally heterogeneous population without inventing a site-distance covariate, a per-event
305 tectonic-regime assignment, and a probability link.

306 4.4 Metrics, calibration, uncertainty, and ranking

307 The primary score is the Brier score [3]; secondary scores are log-loss, AUROC,
308 and calibration intercept/slope [31]. For numerical stability, log-loss clips predicted
309 probabilities symmetrically to $[10^{-15}, 1 - 10^{-15}]$ before taking logarithms. Brier-
310 score uncertainty is quantified by a sequence-cluster bootstrap ($n_{\text{boot}} = 2000$, 95%
311 intervals) that resamples whole sequences rather than events [7], with a minimum-
312 valid-replicate rule. Calibration and feasibility states are *derived*, not hard-coded: a
313 constant-prevalence predictor has no identifiable calibration slope, so the table reports
314 that value as unavailable. Pairwise ranking uses a paired sequence-cluster bootstrap
315 Brier difference with an explicit tie probability; the table reports whether each 95%
316 interval includes zero, together with a tie-aware joint ordering (Section 5.6). Because
317 the intervals use 2,000 Monte Carlo draws, near-zero endpoints are interpreted cau-
318 tiously rather than as stable ranking guarantees. The protocol requested intervals for
319 every metric; only Brier-based intervals were retained, so other metrics are descriptive
320 point estimates (Table 8).

321 4.5 Protected evaluation and reproducibility

322 The official decision-bearing protected evaluation was authorized once by an auto-
323 mated, globally consumed token; the one-shot wrapper refuses a second authorized
324 run. After the result was fixed, the token-free deterministic analytic core—including
325 holdout scoring—was replayed twice for verification, and both replays reproduced the
326 recorded result fingerprint exactly. Thus, single-use refers to result authorization and
327 decision-bearing use, not to the total number of post-result computations. The frozen
328 holdout predictions were also recovered bit-for-bit from the released rows and frozen

best parameters (reported in Section 5.8), under a pinned scientific-Python environment (NumPy, SciPy, scikit-learn, XGBoost; 4, 9, 21, 32). All fits, exclusions, subsets, cutoffs, and pairwise comparisons are reported regardless of outcome; the only permitted omissions are the protocol-defined exclusions above. Computation is local-only under a pinned environment.

Reproducible artwork and generative AI assistance.

All figures were generated with Matplotlib 3.11.1 directly from the frozen analysis assets; no manual image manipulation or AI-generated artwork was used. ChatGPT assisted language editing, formatting, code review, literature discovery, critical consistency review, and revision suggestions. It did not generate data, execute model fits, or create the reported numerical results. The authors independently verified every cited source, computation, interpretation, and assisted change; all scientific judgments and final decisions remained with the authors, who retain full responsibility for the manuscript and released artifacts.

5 Results

All result-bearing values below are generated from the frozen analysis outputs and checked automatically against their source records; no result cell was hand-edited.

5.1 Cohort and roles

Table 2 and Fig. 2 give the cohort accounting. Of 7,520 frozen DYFI records, 2,362 events pass the ten-response eligibility filter (severe prevalence 0.298; 703 severe events), partitioned into 1,573 train, 357 development, and a 430-event temporal hold-out, with two temporal-straddle exclusions. Thus 2,360 of the 2,362 eligible events receive analysis roles; they contain 702 severe events, and the 703rd severe event is one of the two straddle exclusions. Two identifiers prelisted for quarantine are absent

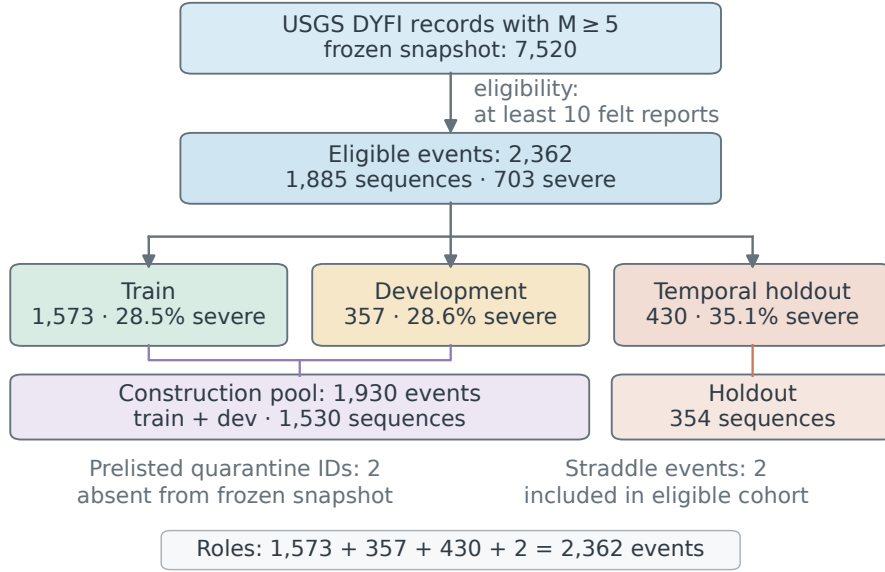


Fig. 2 Cohort construction and role assignment. On the frozen USGS DYFI $M \geq 5$ snapshot. The 7,520 source records yield 2,362 eligible events in 1,885 seismic sequences. Of these, 2,360 are assigned to train ($n = 1,573$), development ($n = 357$), or temporal holdout ($n = 430$); two straddle events are excluded from role assignment

Table 2 Cohort accounting and outcome-independent role assignment

Stage	Events	Severe	Severe rate	Sequences
USGS DYFI records with $M \geq 5$ (frozen snapshot)	7520	–	–	–
Eligible (at least 10 felt reports)	2362	703	0.298	1885
Train	1573	449	0.285	–
Development	357	102	0.286	–
Construction (train+dev)	1930	551	0.285	1530
External temporal holdout	430	151	0.351	354
Temporal-straddle excluded	2	1	–	1

Note. Roles were fixed without using outcomes. Two event identifiers prelisted for quarantine are absent from the frozen snapshot, so the safeguard removes no version 1 record. Severe = 1 when an event's maximum community decimal intensity (CDI) is at least 6. The eligible severe count reconciles as 703 = 449 (train) + 102 (development) + 151 (holdout) + 1 (straddle exclusion). Thus the 2360 role-assigned events contain 702 severe events.

353 from the frozen snapshot, so that safeguard removes no version 1 record. Construc-
 354 tion severe prevalence is 0.285 and holdout prevalence is 0.351; the higher holdout
 355 prevalence reflects the temporal transport shift and is reported, not adjusted.

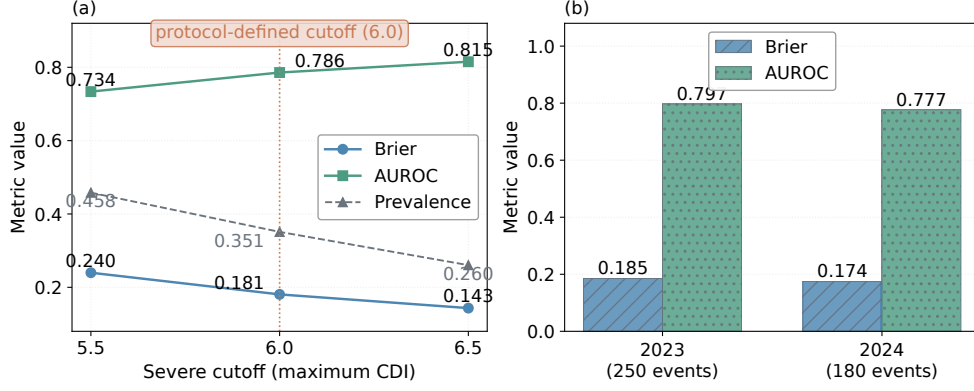


Fig. 3 Endpoint and temporal sensitivity. Results for the source-only logistic regression (B2). (a) Performance as the severe-intensity cutoff varies. (b) Performance within the 2023 ($n = 250$) and 2024 ($n = 180$) portions of the temporal holdout. The cutoff analysis rescores fixed B2 probabilities without refitting. Lower Brier score and higher AUROC are better

Table 3 Endpoint-threshold probe using fixed source-only probabilities

Maximum CDI cutoff	Severe rate	Brier score	AUROC
5.5	0.458	0.240	0.734
6.0 ^a	0.351	0.181	0.786
6.5	0.260	0.143	0.815

Note. ^a Protocol-defined primary cutoff (≥ 6). CDI denotes community decimal intensity. The sensitivity rows relabel and rescore fixed B2 probabilities; B2 is not refit or recalibrated for the alternate cutoffs. In the construction sample, 0.164 of events lie within 0.5 CDI units of the primary cutoff, and the no-skill Brier score is 0.204. Lower Brier score and higher AUROC indicate better performance.

5.2 Difficulty and endpoint sensitivity

Table 3 and the left panel of Fig. 3 report a descriptive fixed-prediction threshold probe. As the severe cut moves from 5.5 to 6.5, holdout prevalence falls smoothly (0.458, 0.351, 0.260) and the source-only logistic-regression Brier score and AUROC move monotonically (Brier 0.240, 0.181, 0.143; AUROC 0.734, 0.786, 0.815). On the construction corpus the near-threshold fraction ($|\text{maxCDI} - 6| < 0.5$) is 0.164 and the no-skill Brier score is 0.204, bounding the difficulty of the primary endpoint. This probe relabels and rescores probabilities fitted for the primary 6.0 endpoint; it does not refit or recalibrate B2 at 5.5 or 6.5 and therefore does not establish endpoint robustness. The protocol-required aggregation-grid sensitivity was not executed and is disclosed as a deviation (Table 8).

367 5.3 Observed split-performance differences

368 The resource’s headline quantity is the optimism of naive random splitting relative
369 to the sequence-grouped protocol, measured for a fixed leakage diagnostic—a random
370 forest with 300 trees and unrestricted depth—under three fold-assignment schemes
371 on the construction corpus (Table 4, Fig. 4). This diagnostic is distinct from the
372 tuned temporal-holdout baseline B4, whose maximum depth is 3. On this $M \geq 5$ one-
373 row-per-event universe the point estimates are small and slightly reversed: sequence
374 grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than
375 random-event splitting, calculated from unrounded scores. The sequence-grouped split
376 therefore scores marginally *better* than the naive random split, not worse. The frozen
377 internal protocol required uncertainty reporting for this contrast but did not define
378 a paired interval estimator; the executed result contains point estimates only. We
379 disclose this unfulfilled element as a deviation and do not invent a post-hoc inter-
380 val. Accordingly, we report the contrast descriptively, not as evidence of equivalence,
381 a true zero effect, leakage, or a win. Its plausible structural context is that 1,885
382 sequences are largely singleton (1,530 in construction), leaving limited within-sequence
383 near-duplicate structure for a random split to leak. This does not prove a small true
384 effect. With one row per event, event grouping imposes the same grouping constraint
385 as random-event assignment, but its deterministic fold allocation differs and there-
386 fore produces different scores. It is reported as an intermediate control. The protocol
387 remains reusable on datasets with more multi-event sequences. The present result nei-
388 ther demonstrates practically important leakage from random splitting nor establishes
389 its absence.

390 5.4 Baseline performance and calibration

391 Table 5 and Fig. 5 give per-baseline holdout performance ($n = 430$) with sequence-
392 cluster bootstrap 95% intervals. The no-skill B0 has Brier 0.232 ($[0.212, 0.253]$); its

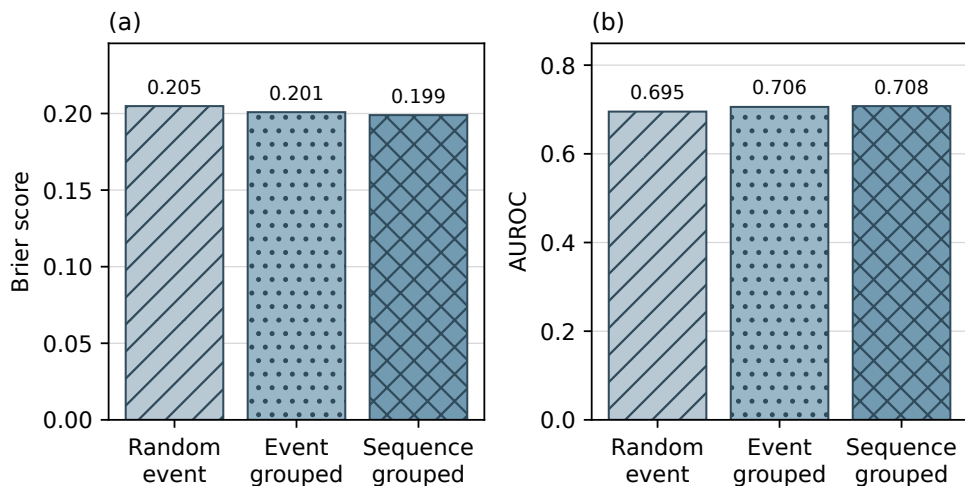


Fig. 4 Split-assignment diagnostic. Five-fold performance of the fixed diagnostic (random forest: 300 trees, unrestricted depth) under random-event, event-grouped, and sequence-grouped fold assignment in the construction sample. Relative to random-event splitting, sequence grouping produced (a) a Brier score 0.0059 lower and (b) an AUROC 0.0124 higher. The diagnostic is distinct from tuned temporal-holdout baseline B4. The small, reversed differences provide no evidence of performance inflation from random splitting in this sample. Both axes start at zero

Table 4 Split-assignment performance of the fixed diagnostic

Fold assignment	Brier score	AUROC	Log-loss	Folds
Random-event (naive)	0.2048	0.6952	0.9640	5
Event-grouped (control)	0.2009	0.7058	0.9223	5
Sequence-grouped (primary)	0.1990	0.7076	0.9709	5

Note. The diagnostic is distinct from the tuned temporal-holdout baseline B4 (depth 3). Using unrounded values, sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting. The frozen internal protocol required uncertainty but did not define the paired interval estimator; the executed report retained point estimates only. This deviation is disclosed, and no post-hoc interval or equivalence claim is added. Because there is one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation is different and can produce different scores.

393 calibration slope cannot be estimated because it assigns the same probability to every
394 event. The learned baselines have lower Brier (B1 0.198, B2 0.181, B4 0.187, B5 0.181)
395 with overlapping intervals; calibration intercept/slope pairs are finite and positive. No
396 baseline is claimed to win; the table reports proper scores, uncertainty, and derived
397 calibration states only.

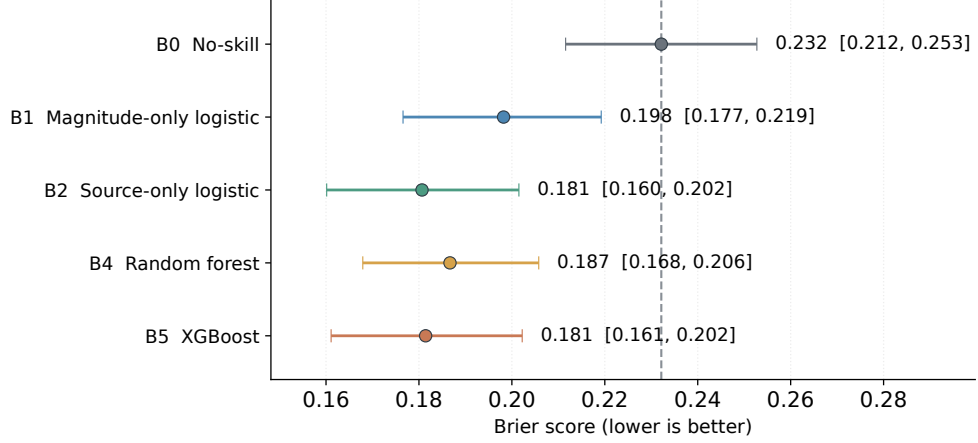


Fig. 5 Temporal-holdout baseline performance. Brier scores and 95% sequence-cluster bootstrap confidence intervals for the five baselines on the temporal holdout ($n = 430$). Lower values are better; the vertical dashed line marks the no-skill B0 score. B3 is omitted because event year is outside the declared source-only temporal-holdout contract (Table 8)

Table 5 Temporal-holdout performance of fixed baseline models

Baseline	Brier	95% CI	Log-loss	AUROC	Calibration ^a	Accuracy
B0 (no-skill)	0.232	[0.212, 0.253]	0.658	0.500	INFEASIBLE	0.65
B1 (magnitude-only logistic)	0.198	[0.177, 0.219]	0.583	0.719	0.32 / 1.02	0.70
B2 (source-only logistic)	0.181	[0.160, 0.202]	0.536	0.786	0.64 / 1.29	0.73
B4 (random forest)	0.187	[0.168, 0.206]	0.553	0.777	0.74 / 1.48	0.73
B5 (XGBoost)	0.181	[0.161, 0.202]	0.541	0.783	0.42 / 1.07	0.74

Note. ^a Intercept / slope. These values cannot be estimated for B0 because it assigns the same probability to every event. B3 includes event year and is therefore omitted from the temporal holdout (Table 8).

Accuracy uses a fixed probability cutoff of 0.5 that was not selected on the holdout. Log-loss clips probabilities to $[10^{-15}, 1 - 10^{-15}]$. Lower Brier score and log-loss are better; higher AUROC and accuracy are better. The two lowest-Brier models cannot be distinguished in their direct paired comparison.

5.5 Worked candidate qualification

Table 6 applies the released engineering decision rule to the six fixed candidates. B3 is outside the source-only contract because it contains event year; this is a conservative scope exclusion, not demonstrated label leakage. B1, B2, B4, and B5 pass the audit and each has a paired Brier interval below the no-skill B0 reference. The contract therefore supports their improvement over B0. It does not support naming B2 or B5 as the winner: their direct paired interval includes zero. This worked example demonstrates the benchmark’s intended use—admit only valid candidates, support improvements with uncertainty, and leave an unresolved comparison unresolved.

Table 6 Worked candidate-model qualification using the frozen temporal holdout

Candidate	Audit gate	Holdout [95% CI]	Brier	Brier difference vs B0 [95% CI]	Contract decision
B0 no-skill	Pass (reference)	0.232 0.253]	[0.212,	—	Reference only
B1 magnitude-only logistic	Pass	0.198 0.219]	[0.177,	-0.0339 [-0.0511, -0.0171]	Supported vs B0
B2 source-only logistic	Pass	0.181 0.202]	[0.160,	-0.0515 [-0.0681, -0.0342]	Supported vs B0
B4 random forest	Pass	0.187 0.206]	[0.168,	-0.0455 [-0.0598, -0.0307]	Supported vs B0
B5 XGBoost	Pass	0.181 0.202]	[0.161,	-0.0507 [-0.0682, -0.0319]	Supported vs B0
B3 expanded-metadata logistic	Out of contract	—	—	—	Year excluded by source-only schema

Decision rule. A candidate is first admitted by the schema, proxy, role, and temporal audits. A lower paired Brier interval that excludes zero supports improvement over the declared reference; otherwise the comparison remains unresolved. All four learned baselines improve over B0. B2 versus B5 remains unresolved (B2–B5 -0.0008, 95% CI [-0.0078, 0.0068]), so the contract prevents a false winner declaration.

5.6 Pairwise ranking stability

Table 7 reports all ten pairwise Brier-score comparisons. Every learned baseline performs better than the no-skill reference B0 (all four intervals exclude zero). Among the learned models, the B2–B4 and B2–B5 comparisons have intervals that include zero. The remaining learned-model intervals exclude zero, although the B4–B5 lower endpoint (0.0003) is near the Monte Carlo boundary. In the joint bootstrap ordering, B2 has the highest probability of being uniquely best (0.613), followed by B5 (0.387), but the direct B2–B5 comparison remains uncertain. These results therefore do not justify naming a winning model.

The frozen internal protocol allowed an “equivalent” verdict but did not specify a positive numerical equivalence margin; its resolution floor was zero. We therefore do not invent a post-hoc margin or label any models equivalent. Instead, we report each point difference, 95% interval, and whether that interval includes zero. Table 8 records this interpretive restriction, which changes no result.

Table 7 Paired Brier-score comparisons among admissible baselines

Model A	Model B	A − B	95% CI	$P(A \text{ better})$	CI relation
B0	B1	0.0339	[0.0171, 0.0511]	0.000	excludes 0
B0	B2	0.0515	[0.0342, 0.0681]	0.000	excludes 0
B0	B4	0.0455	[0.0307, 0.0598]	0.000	excludes 0
B0	B5	0.0507	[0.0319, 0.0682]	0.000	excludes 0
B1	B2	0.0176	[0.0101, 0.0249]	0.000	excludes 0
B1	B4	0.0115	[0.0023, 0.0205]	0.009	excludes 0
B1	B5	0.0168	[0.0055, 0.0273]	0.001	excludes 0
B2	B4	-0.0060	[-0.0118, 0.0002]	0.970	includes 0
B2	B5	-0.0008	[-0.0078, 0.0068]	0.613	includes 0
B4	B5	0.0052	[0.0003, 0.0099]	0.017	excludes 0

Note. Lower Brier score is better, so a negative A − B difference favors model A. Intervals use 2,000 Monte Carlo bootstrap draws; near-zero endpoints are interpreted cautiously and not as stable ranking guarantees. Because no numerical equivalence margin was specified before analysis, intervals including zero are not labelled equivalent. B0 is the no-skill reference; B1 is magnitude-only logistic regression; B2 is source-only logistic regression; B4 is random forest; and B5 is XGBoost.

5.7 Temporal and geographic subgroup summaries

Table 9 and the right panel of Fig. 3 report the temporal and geographic subgroup summaries for the source-only logistic regression within the temporal holdout. The two holdout years are internally consistent (2023: $n = 250$, Brier 0.185, AUROC 0.797; 2024: $n = 180$, Brier 0.174, AUROC 0.777). The geographic subgroup summary spans 82 occupied 10° cells (per-cell Brier 0.002–0.825, median 0.151). These values stratify the same 430 temporal-holdout predictions; they are not a leave-region-out evaluation and support no geographic or cross-system generalization claim.

5.8 Exclusions, deviations, and reproducibility

Table 8 explains the protocol decisions most relevant to interpretation: excluded comparators and records, the unexecuted geographic transport branch, the unfulfilled split-contrast interval, and the absence of an equivalence claim. The tagged public release contains the complete SHA-256 fingerprints, software environment, and machine-readable verification details. Its public-release manifest fingerprints every distributed file. Three original execution-bookkeeping records are self-referential—they cannot contain their own final fingerprints—and are therefore reported separately

Table 8 Protocol exclusions, deviations, and interpretation boundaries

Decision	Treatment	Rationale
B3 expanded-metadata model	Omitted from temporal holdout	Includes event year, which reveals the temporal evaluation period.
Published DYFI intensity equation	Omitted as comparator	Needs unavailable site distance, tectonic-region inputs, and probability conversion.
Two prelisted quarantine IDs	No effect in version 1	Both are absent from the frozen snapshot; no released record is excluded.
Geographic transport evaluation	Not executed	Only temporal-holdout geographic subgroups are reported; no leave-region-out or generalization claim.
Aggregation-grid sensitivity	Not executed	Required but absent; only threshold sensitivity of fixed predictions is reported.
Uncertainty coverage	Partial deviation	The protocol requested intervals for every metric and the split contrast; only Brier-based intervals were retained.
Equivalence interpretation	Descriptive only	No numerical margin was specified; intervals are reported without equivalence labels.

Note. These decisions were recorded before, or disclosed with, the analysis. No result, subset, cutoff, or comparison was omitted because of its observed outcome.

Table 9 Temporal and geographic summaries within the temporal holdout

Subset	Events	Brier score	AUROC
Temporal 2023	250	0.185	0.797
Temporal 2024	180	0.174	0.777
Geographic cells within temporal holdout (82)	430	0.151 (med.)	–

Note. The geographic row partitions the same 430 temporal-holdout predictions into 82 occupied 10° cells. It is a descriptive subgroup summary, not a leave-region-out or external geographic evaluation. The entry is the median cell-level Brier score; cell values range from 0.002 to 0.825 and are included in the public release.

rather than treated as analysis-file mismatches. A machine-readable corrections
 overlay deprecates three legacy semantic fields (protected-role status, the generic
 split-interpretation template, and event-group wording) and provides authoritative
 replacements without changing any numerical result.

441 6 Discussion

442 *Engineering use and decision consequence.*

443 The resource supports a transparent retrospective audit of an event-level classifier.
444 The user does not choose a convenient split or reconstruct a moving dataset; the
445 declared candidate is evaluated against a released contract. The audit rejects forbid-
446 den outcome-derived inputs, the role manifest prevents events linked by the declared
447 heuristic from crossing roles, and the paired Brier interval determines whether improve-
448 ment over a reference is supported. The worked example changes two decisions: B3 is
449 outside the source-only contract, while B2 and B5 are both supported over no-skill
450 but cannot be ranked decisively. Without the contract, the same point scores invite
451 an unsupported winner claim. Because all labels are public, a developer can tune to
452 this release; prospective or confirmatory qualification requires a fresh hidden holdout
453 or independent evaluation service.

454 *Interpreting the split contrast.*

455 The observed point estimate is small and slightly reversed. In these folds the naive
456 random split did not inflate the diagnostic score, but the missing interval pre-
457 vents an equivalence, zero-effect, or “little practical effect” inference. The protocol
458 remains useful for datasets containing more multi-event sequences; this result neither
459 demonstrates leakage nor supports a win.

460 *No winning model.*

461 Every learned baseline separates from the no-skill reference. However, the direct B2–
462 B5 comparison between the two lowest-Brier models is unresolved, as is B2–B4, so
463 the results do not establish a decisive winner. Consistent with the resource’s scope,
464 we claim no winning model and no operational, real-time, lead-time, cross-system, or
465 causal benefit.

466 7 Limitations and threats to validity

467 *Outcome and catalogue limitations.*

468 The endpoint is magnitude-dominated, CDI remains sensitive to responder coverage,
469 and magnitude and depth are frozen preferred catalogue values rather than guaranteed
470 first-alert values. The ten-response rule sets an eligibility and reliability floor but
471 selects on post-event participation and does not remove ascertainment; response count
472 is used only for eligibility. B1 provides the simple magnitude comparator.

473 *Unexecuted sensitivity and transport branches.*

474 The geographic values are temporal-holdout subgroups, not external transport evi-
475 dence. The protocol-defined geographic evaluation and aggregation-grid sensitivity
476 were not executed, and no alternative to the fixed 100-km/30-day sequence defini-
477 tion was tested. The rule guarantees co-assignment only for events it links; it does
478 not guarantee identification of every physical aftershock relationship. Consequently,
479 no geographic, aggregation, sequence-definition, or complete aftershock-isolation
480 robustness claim is made.

481 *Inference and held-out semantics.*

482 The protocol-required split-contrast uncertainty interval was not retained, so that
483 result remains descriptive with no equivalence or zero-effect inference. Non-Brier
484 scores remain descriptive point estimates. Pairwise intervals use 2,000 Monte Carlo
485 draws and near-zero endpoints are interpreted cautiously. The temporal holdout was
486 excluded from model development, but aggregate feasibility was previously known; it
487 is neither “untouched” nor confirmatory.

488 *Public release.*

489 Version 1 of the data, code, frozen results, and reproduction instructions
490 is available in a tagged public repository at <https://github.com/akssha74/>

491 [dyfi-benchmark-reliability](#). The release retains complete digital fingerprints so that
492 later revisions remain distinguishable from the version evaluated here.

493 8 Conclusion

494 Reliable engineering model selection requires more than reporting the lowest point
495 score. The released DYFI benchmark makes a candidate pass an auditable schema
496 and role contract, evaluates it on a fixed temporal holdout, and attaches paired uncer-
497 tainty to the comparison decision. In the worked qualification, four learned candidates
498 improve over no-skill, B3 is outside the declared source-only contract, and the two
499 lowest-Brier models remain unresolved; the contract therefore prevents an out-of-scope
500 comparison and a false winner declaration. The random-versus-sequence diagnostic is
501 small and reversed in this sparse-sequence corpus, so it supplies no evidence of random-
502 split optimism. All results reconstruct from the frozen release. The contribution is a
503 reusable retrospective test harness for trustworthy model comparison. Confirmatory
504 use requires a fresh hidden holdout or independent evaluation service; the release is
505 not a real-time warning, causal model, or operational decision system.

506 Declarations

- 507 • **Funding.** This research did not receive any specific grant from funding agencies in
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- 509 • **Competing interests.** The authors have no relevant financial or non-financial
510 interests to disclose.
- 511 • **Ethics approval.** Not applicable. The study analyzes public, aggregate event-level
512 USGS products rather than individual questionnaire responses; it involved no inter-
513 action with participants, identifiable data, human-subject collection, animals, or
514 biological material.
- 515 • **Consent to participate.** Not applicable; no participants were recruited.

- 516 • **Consent for publication.** Not applicable.
- 517 • **Availability of data and materials.** Version 1 data, results, manifests,
 518 and instructions are available under CC0 1.0 at [https://github.com/akssha74/](https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.1.1)
 519 [dyfi-benchmark-reliability/releases/tag/v1.1.1](https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.1.1). Source data are USGS DYFI and
 520 ANSS ComCat (public domain; [10.5066/F7J101C8](https://doi.org/10.5066/F7J101C8) and [10.5066/F7MS3QZH](https://doi.org/10.5066/F7MS3QZH)) and
 521 are reconstructable via the released fetch-and-verify source-locator manifest. No
 522 respondent-level personally identifying information is redistributed.
- 523 • **Availability of code.** Project: DYFI Benchmark Reliability. Homepage: [https://](https://github.com/akssha74/dyfi-benchmark-reliability)
 524 github.com/akssha74/dyfi-benchmark-reliability. Archived version: release v1.1.1.
 525 Operating systems: platform independent. Programming language: Python. Licence:
 526 MIT. Restrictions on non-academic use: none. Full prediction reconstruction and
 527 model tests use the separately pinned scientific-Python environment.
- 528 • **Author contributions.** A.S. conceived and designed the benchmark protocol,
 529 developed the software, curated the data, performed the analyses, validated the
 530 results, and drafted the manuscript. L.P. contributed to study design and inter-
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